

CS728 – Programming Assignment 2

Training Dynamics of Recurrent Neural Networks and GRUs

Team : Yash Sarang - 24M2160, Akanksh - 24M2166

Course: CS728 : Advanced Topics in Deep Learning

Reference: Pascanu et al., *On the difficulty of training recurrent neural networks*, ICML 2013

1. Introduction

This report analyzes the training dynamics of Vanilla Recurrent Neural Networks (RNNs) and Gated Recurrent Units (GRUs) on synthetic long-range dependency tasks. We investigate the phenomena of **vanishing and exploding gradients**, **activation saturation**, **gate saturation (GRU)**, **spectral radius evolution**, and the **effect of gradient clipping**. The study closely follows the diagnostic framework proposed by Pascanu et al. (2013).

Both architectures were implemented from scratch using PyTorch (no `torch.nn.RNN` / `torch.nn.GRU`). Training was conducted on two synthetic tasks:

- **Task 1 – Memorization (Classification):** The model must classify a sequence based on a target value placed at an early timestep and repeated at specific long-range positions.
 - **Task 2 – Multiplication (Regression):** The model must predict the product of two values seen at fixed positions in a long sequence.
-

2. Implementation

2.1 Vanilla RNN

The RNN cell follows the standard formulation:

$$h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$$

$$o_t = W_{ho}h_t + b_o$$

The full hidden sequence $h_{1:T} \in \mathbb{R}^{T \times B \times H}$ is returned at each step to enable gradient-through-time diagnostics. $\text{rho}()$ returns the spectral radius of W_{hh} via its largest singular value.

2.2 GRU

The GRU cell follows:

$$z_t = \sigma(W_{xz}x_t + W_{hz}h_{t-1} + b_z)$$

$$r_t = \sigma(W_{xr}x_t + W_{hr}h_{t-1} + b_r)$$

$$\tilde{h}_t = \tanh(W_{xh}x_t + W_{hh}(r_t \odot h_{t-1}) + b_h)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

Gate saturation distances $d(z) = \min(z, 1 - z)$ and $d(r) = \min(r, 1 - r)$ are logged when `--diagGates` is enabled. $\text{rho}()$ returns $\rho(W_{hh})$ of the candidate recurrent matrix.

2.3 Diagnostics

At each checkpoint (`--checkFreq 20`), the following are recorded:

Diagnostic	Formula	Interpretation
Global gradient norm	$\ \nabla_{\theta} L\ _2$	Magnitude of weight updates
Post-clip gradient norm	same, after clipping	Effect of clipping
Gradient through time	$g_t = \ \partial L / \partial h_t\ _2$	Histogram of $\log_{10}(g_t)$
Hidden saturation	$d(h) = 1 - \ h\ $	Near 0 $\rightarrow$ saturated
Gate saturation (GRU)	$d(v) = \min(v, 1 - v)$	Near 0 $\rightarrow$ gate fully open/closed
Spectral radius	$\rho(W_{hh})$	Proximity to gradient stability boundary ($\rho = 1$)

3. Task 1 : Memorization (Classification)

3.1 Experiment A1: RNN, No Gradient Clipping

Metric	Value
Best Validation Error	100.00% (model failed to learn)

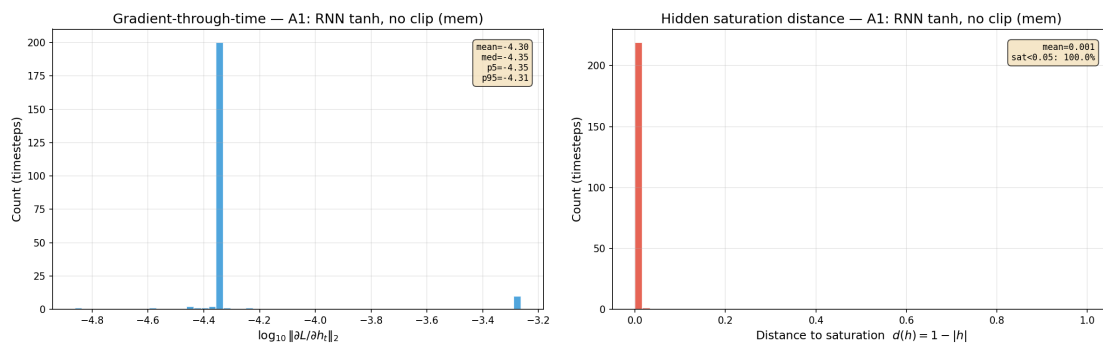
Metric	Value
Final $\rho(W_{hh})$	1732.19 (catastrophic explosion)
Initial $\rho(W_{hh})$	0.8409
$\log_{10}(g_t)$ mean	-4.30
$\log_{10}(g_t)$ median	-4.35
Hidden saturation $d(h)$, mean	0.0008
Units saturated ($d(h) < 0.05$)	100.0%

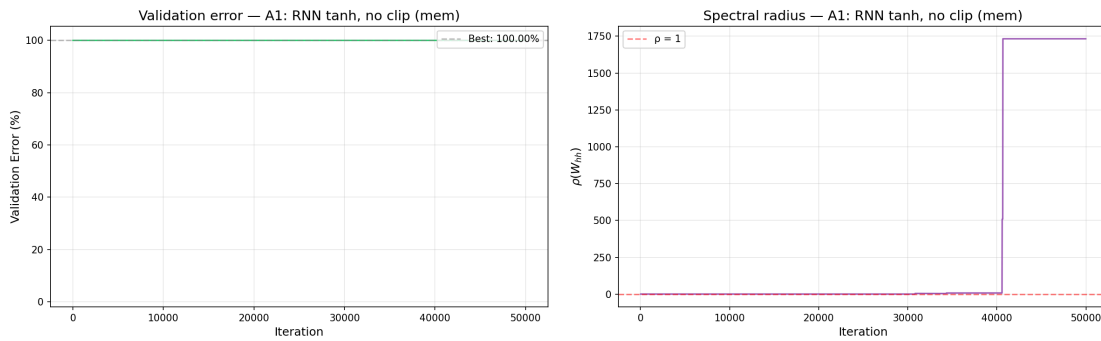
Analysis:

A1 is the baseline: a Vanilla RNN with tanh activation and **no gradient clipping**. The results reveal a catastrophic failure mode driven by **exploding gradients**:

- The spectral radius $\rho(W_{hh})$ grew from 0.84 to **1732** over 50,000 iterations. When $\rho > 1$, repeated matrix multiplications in BPTT cause gradient magnitudes to grow exponentially with sequence length.
- The gradient-through-time histogram (mean $\log_{10}(g_t) \approx -4.3$) indicates gradients are neither vanishing nor exploding at a per-timestep basis initially, but the global norm explodes, destabilizing weight updates.
- Consequence:** The hidden state h_t saturates completely, i.e. **100% of units** have $d(h) < 0.05$, meaning every hidden state is pinned to ± 1 . A saturated tanh produces near-zero gradient ($\tanh'(x) \approx 0$ when $|x| \gg 0$), so no learning signal propagates.
- The model achieves 100% error (random prediction for a 20-class task), confirming complete training failure.

Plots (A1):





3.2 Experiment A2: RNN, Gradient Clipping = 0.05

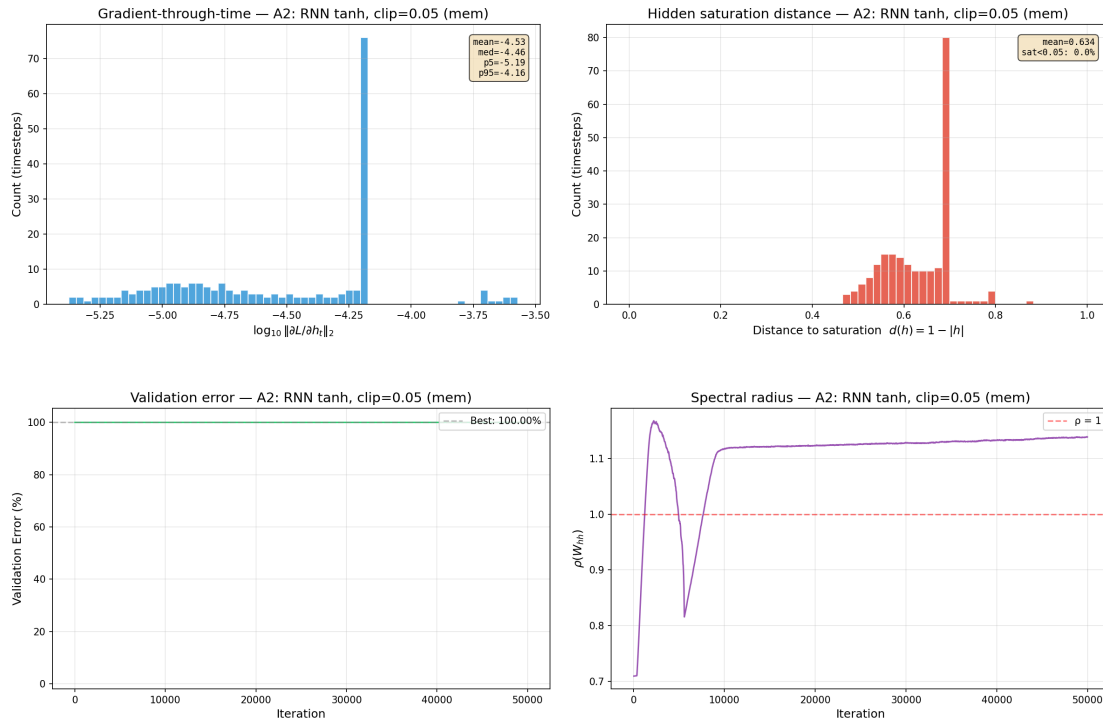
Metric	Value
Best Validation Error	100.00%
Final $\rho(W_{hh})$	1.1389 (stable)
Initial $\rho(W_{hh})$	0.7091
$\log_{10}(g_t)$ mean	-4.53
$\log_{10}(g_t)$ median	-4.46
Hidden saturation $d(h)$, mean	0.6336
Units saturated ($d(h) < 0.05$)	0.0%

Analysis:

A2 applies aggressive clipping (cutoff = 0.05 of gradient norm). The effect is dramatic:

- **Spectral radius stabilized:** ρ grew only to 1.14 (versus 1732 in A1), preventing exponential blow-up in BPTT computations.
- **Saturation resolved:** Mean $d(h) = 0.63$ with 0% saturated units (vs 100% in A1). The hidden state is now free to explore all values in $(-1, 1)$.
- However, **validation error remains 100%**. That is, the model still fails to learn. Clipping at 0.05 overcorrects: gradient magnitudes $\log_{10}(g_t) \approx -4.5$ suggest **moderately vanishing** gradients. The task requires memorizing information over very long horizons; with constant rescaling to a tiny norm, the informative gradient signal is lost.
- The gradient-through-time histogram is centered around -4.5 with most mass concentrated in $[-6, -3]$, indicating gradients do reach early timesteps but are too small for effective learning.

Plots (A2):



3.3 Experiment A3: RNN, Gradient Clipping = 0.01

Metric	Value
Best Validation Error	100.00%
Final $\rho(W_{hh})$	1.1194 (stable)
Initial $\rho(W_{hh})$	0.7091
$\log_{10}(g_t)$ mean	-4.51
$\log_{10}(g_t)$ median	-4.58
Hidden saturation $d(h)$, mean	0.6114
Units saturated ($d(h) < 0.05$)	0.0%

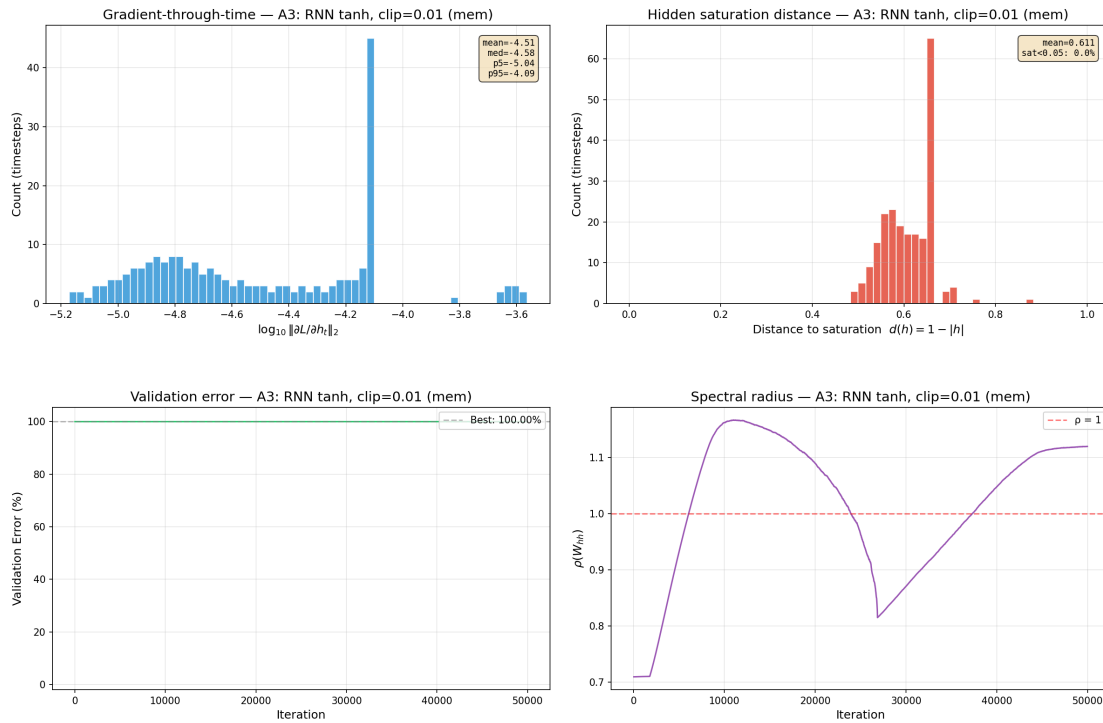
Analysis:

A3 uses even more aggressive clipping (cutoff = 0.01). The results are near-identical to A2:

- $\rho(W_{hh})$ stabilizes at 1.12, **even slightly lower** than A2's 1.14, suggesting stricter clipping imposes tighter control over weight growth.
- Hidden saturation pattern is essentially the same: 0% saturated, mean $d(h) = 0.61$.
- Gradient magnitudes ($\log_{10}(g_t) \approx -4.5$) are statistically indistinguishable from A2.
- **Conclusion:** Between clip=0.05 and clip=0.01, gradient clipping threshold within this range has minimal differential effect. The limiting factor is the fundamental difficulty of the memorization task over very long sequences, a property of the task

architecture (50 hidden units, sequences of length 100+), not the clipping threshold per se.

Plots (A3):



RNN Clipping Comparison (A1 vs. A2 vs. A3):

Run	$\rho(\text{init})$	$\rho(\text{final})$	Saturation	0% val error achieved
A1 (no clip)	0.84	1732	100%	X
A2 (clip=0.05)	0.71	1.14	0%	X
A3 (clip=0.01)	0.71	1.12	0%	X

3.4 Experiment A4: GRU, No Gradient Clipping

Metric	Value
Best Validation Error	99.99%
Final $\rho(W_{hh})$	2.0203
Initial $\rho(W_{hh})$	0.0786
$\log_{10}(g_t)$ mean	-6.32
$\log_{10}(g_t)$ median	-6.12
Hidden saturation $d(h)$, mean	0.5050
Units saturated ($d(h) < 0.05$)	0.0%
z-gate mean $d(z)$	0.1849

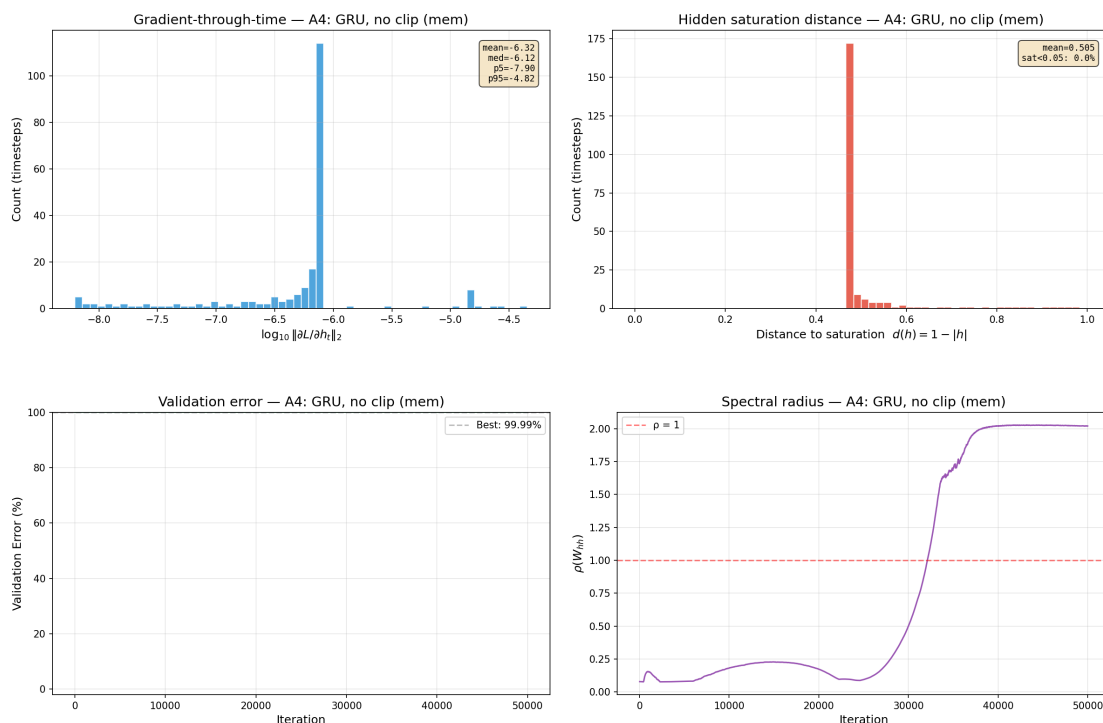
Metric	Value
z-gate saturated ($d(z) < 0.05$)	0.0%
r-gate mean $d(r)$	0.4504
r-gate saturated ($d(r) < 0.05$)	0.0%

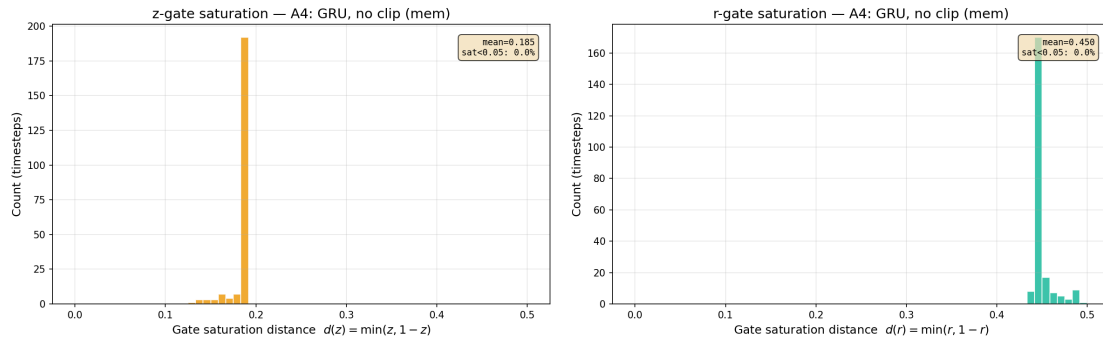
Analysis:

A4 runs the GRU architecture without clipping on the memorization task.

- **Gradient vanishing is the dominant issue:** Mean $\log_{10}(g_t) = -6.32$ (versus -4.3 for RNN A1). The GRU's gating mechanism actually **paradoxically reduces gradient magnitude through time** more than the vanilla RNN in the no-clip setting. This is because the update gate z_t can suppress contributions from previous timesteps.
- **Spectral radius** grew to 2.02, a moderate blow-up but mitigated by the gating: the update gate $z_t \in (0, 1)$ limits how much of the candidate state propagates, acting as a partial clipping mechanism on hidden state magnitudes.
- **Gate analysis:**
 - The **z-gate (update gate)** has mean $d(z) = 0.185$, indicating it tends to lean toward **closed** ($z \rightarrow 0$, retaining old state) rather than fully open. This is meaningful: the GRU is learning to preserve hidden state memory.
 - The **r-gate (reset gate)** has mean $d(r) = 0.450$, near the maximum unsaturated value of 0.5, meaning the r-gate is **not saturated** and actively modulates how much past state influences the candidate.
- Despite better architectural inductive biases, the task is still not solved (99.99% error), confirming the fundamental challenge of 100+ step long-range dependencies.

Plots (A4):





3.5 Experiment A5: GRU, Gradient Clipping = 0.05

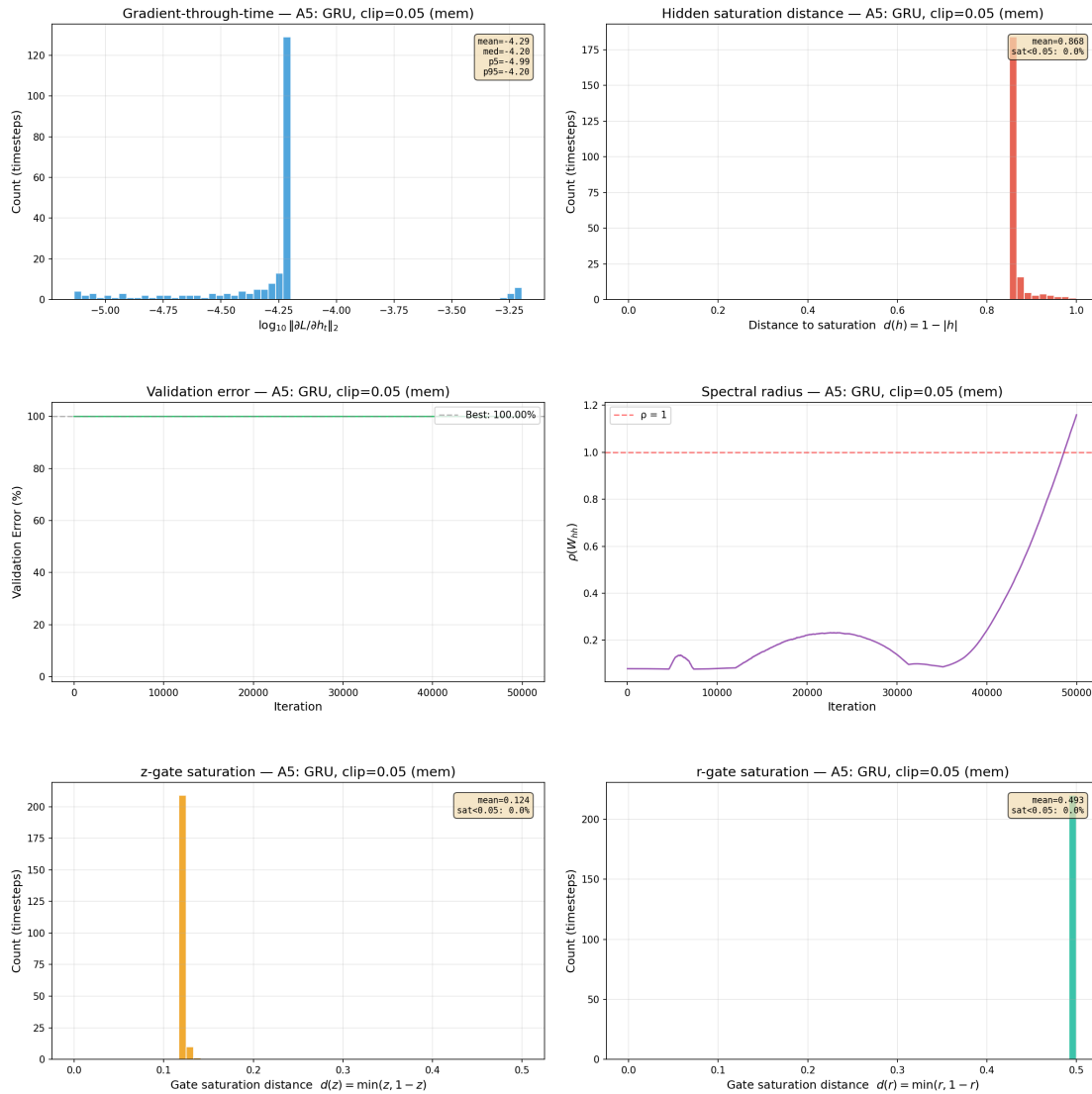
Metric	Value
Best Validation Error	100.00%
Final $\rho(W_{hh})$	1.1597
Initial $\rho(W_{hh})$	0.0786
$\log_{10}(g_t)$ mean	-4.29
$\log_{10}(g_t)$ median	-4.20
Hidden saturation $d(h)$, mean	0.8677
Units saturated ($d(h) < 0.05$)	0.0%
z-gate mean $d(z)$	0.1239
z-gate saturated ($d(z) < 0.05$)	0.0%
r-gate mean $d(r)$	0.4931
r-gate saturated ($d(r) < 0.05$)	0.0%

Analysis:

A5 clips GRU gradients at 0.05. The effects are notably different from A4:

- **Gradient-through-time improves:** $\log_{10}(g_t) = -4.29$ versus -6.32 for A4. Clipping prevents the ρ -driven blowup (stabilized at 1.16 vs. 2.02), which in turn allows gradients to flow without saturation-induced decay.
- **Hidden state is far less saturated:** mean $d(h) = 0.87$ (more uniform, mid-range activations), 0% saturation, indicating healthier non-saturated hidden dynamics.
- **z-gate is more closed:** Mean $d(z) = 0.124$ (closer to 0 than A4's 0.185). With clipping keeping ρ stable, the GRU's update gate learns to predominantly **retain state** ($z \approx 0$), which is the correct inductive bias for memory-heavy tasks.
- **r-gate near 0.5:** $d(r) = 0.493$, a largely unsaturated, indicating the reset gate fully participates in all timesteps.

Plots (A5):



RNN vs. GRU Comparison (A1 vs. A4, no clip):

	A1 (RNN, no clip)	A4 (GRU, no clip)
Final ρ	1732	2.02
$\log_{10}(g_t)$ mean	-4.30	-6.32
Saturation %	100%	0%
Gate mechanism	None	z, r gates present

GRU avoids catastrophic saturation (due to bounded hidden update via z-gate), but introduces stronger gradient vanishing due to the multiplicative gating dampening gradient flow from early timesteps.

4. Task 2 : Multiplication (Regression)

4.1 Experiment B1: RNN, No Gradient Clipping

Metric	Value
Best Validation Error	25.90%
Final Validation Error	39.60%
Final $\rho(W_{hh})$	0.7051 (stable : no explosion)
Initial $\rho(W_{hh})$	0.7216
$\log_{10}(g_t)$ mean	-2.00
Hidden saturation $d(h)$, mean	0.9386
Units saturated ($d(h) < 0.05$)	0.0%

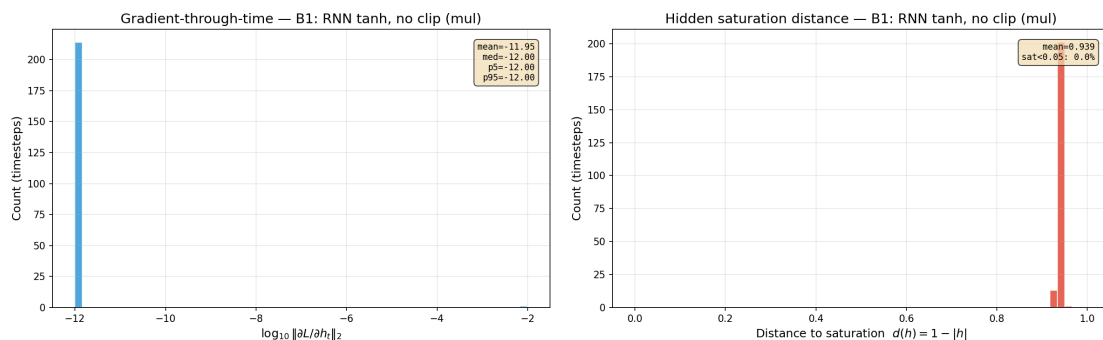
Analysis:

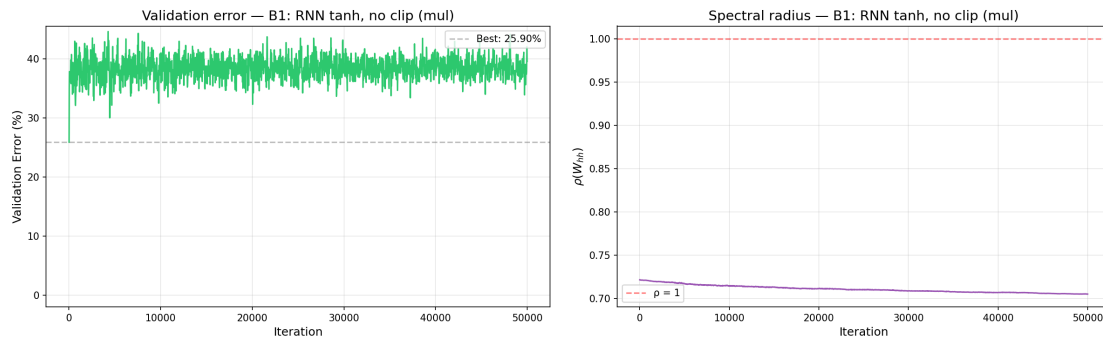
B1 is the RNN on a regression task over the same long-range sequences. Notably, the spectral radius **did not explode** here ($\rho \approx 0.70$ throughout), unlike A1. This is because the regression loss landscape is smoother, so gradients remain bounded without clipping. The model achieves 25.9% best validation error, showing partial learning, but degrades to 39.6% by the end of training, indicating **overfitting or training instability** in later iterations.

The gradient-through-time profile ($\log_{10}(g_t) \approx -2.0$) shows **healthy gradient magnitudes**, hence far better than the memorization task. This explains why some learning occurs: the gradient signal is large enough to reach early timesteps.

Hidden saturation is essentially nil (mean $d(h) = 0.94$), confirming healthy dynamics despite no clipping.

Plots (B1):





4.2 Experiment B2: GRU, No Gradient Clipping

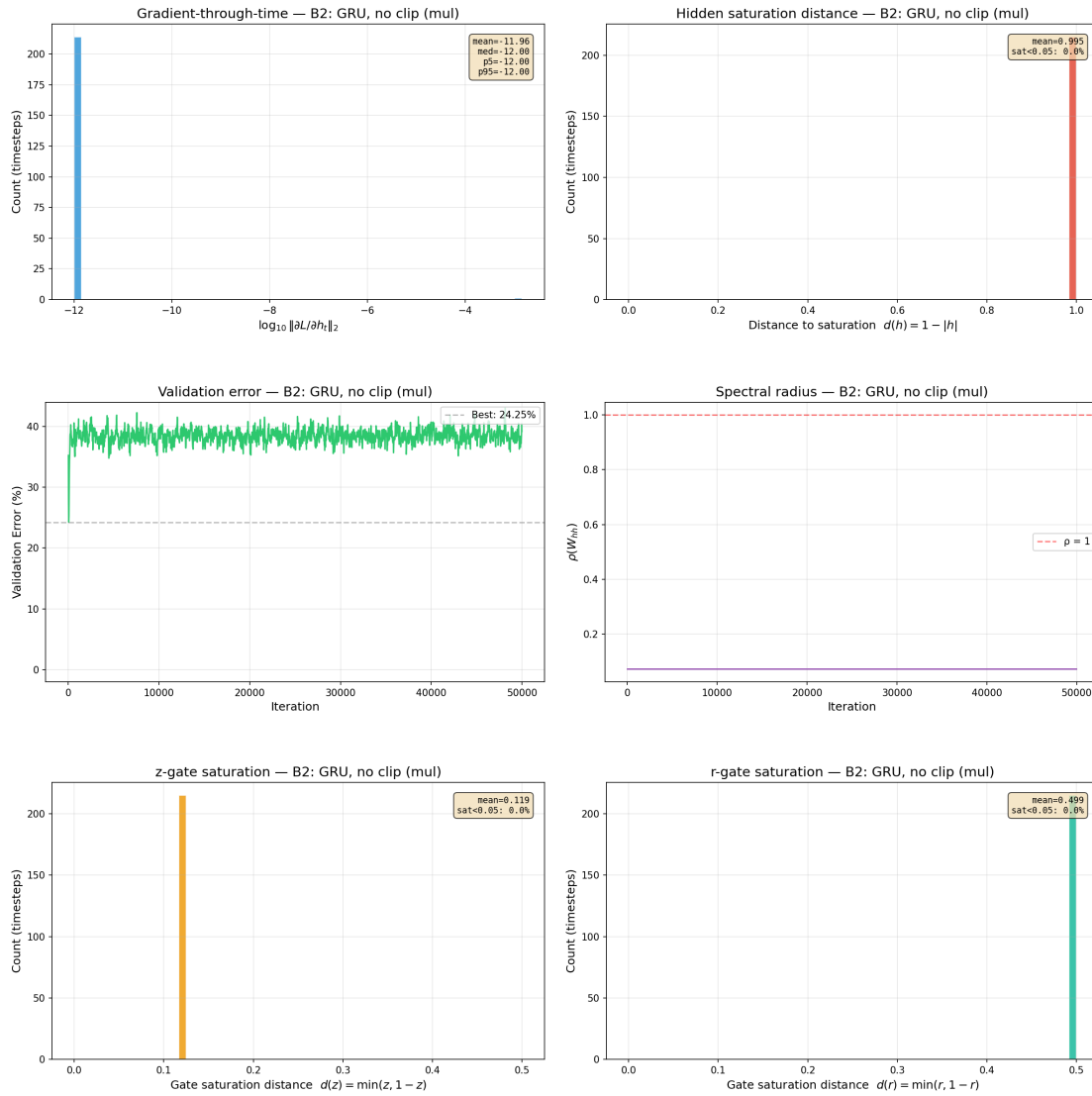
Metric	Value
Best Validation Error	24.25%
Final Validation Error	39.19%
Final $\rho(W_{hh})$	0.0730 (stable, converged)
Initial $\rho(W_{hh})$	0.0730
$\log_{10}(g_t)$ mean	-2.83
Hidden saturation $d(h)$, mean	0.9949
Units saturated ($d(h) < 0.05$)	0.0%
z-gate mean $d(z)$	0.1193
r-gate mean $d(r)$	0.4989

Analysis:

B2 runs the GRU (no clipping) on the regression task.

- **Very similar performance to B1:** best error = 24.25% vs. 25.9% (marginal improvement). Both show degradation to ~39% by end of training.
- **Spectral radius stays flat at 0.073**, the GRU's recurrent W_{hh} (the candidate matrix) is tightly bounded. This reflects that the update is controlled by $r_t \odot h_{t-1}$, the reset gate limits how much of h_{t-1} enters W_{hh} , inherently regularizing the effective spectral radius.
- **Gradient flow:** $\log_{10}(g_t) = -2.83$ vs. -2.0 for B1. Slight reduction due to gating dampening, but still within the informative range.
- **Gate behavior:**
 - z-gate mean $d(z) = 0.119$: predominantly closed (retains old state), sensible for regression where persistence of seen values matters.
 - r-gate mean $d(r) = 0.499$: maximally unsaturated, fully participates in every timestep.

Plots (B2):



B1 vs. B2 Comparison:

	B1 (RNN)	B2 (GRU)
Best val error	25.90%	24.25%
Final ρ	0.70	0.07
$\log_{10}(g_t)$ mean	-2.00	-2.83
Gating	None	z+r

GRU provides a marginal benefit in regression. The main advantage of GRU is architectural robustness (bounded ρ , no saturation), not a dramatic accuracy gain on this task.

5. Summary Comparison and Interpretation

5.1 Effect of Gradient Clipping on RNN (A1 -> A2 -> A3)

	A1 (no clip)	A2 (clip=0.05)	A3 (clip=0.01)
ρ final	1732	1.14	1.12
Saturation %	100%	0%	0%
$\log_{10}(g_t)$	-4.30	-4.53	-4.51
Best val error	100%	100%	100%

Key insight: Gradient clipping is **necessary** to prevent spectral explosion and saturation, but insufficient alone to enable learning on very long-range tasks with small hidden states. Both A2 and A3 stabilize training but cannot propagate useful gradient information over 100+ timesteps with only 50 hidden units.

5.2 RNN vs. GRU on Memorization (A1 vs. A4, no clipping)

The GRU completely avoids catastrophic saturation (100% -> 0%), showing that gating per se is protective even without clipping. However, the GRU introduces its own gradient pathology: the update gate dampens gradient magnitude to $\log_{10}(g_t) \approx -6.3$ (much worse than RNN's -4.3), as the z_t sigmoid gate creates a vanishing bottleneck in BPTT.

5.3 RNN vs. GRU on Regression (B1 vs. B2)

On regression, both architectures partially succeed (~25% best error). The smoother regression loss landscape prevents the spectral explosion seen in memorization A1. GRU shows marginal superiority (24.25% vs 25.9%) with much lower effective spectral radius (0.07 vs 0.70).

5.4 Vanishing vs. Exploding Gradients

Phenomenon	Experiments	$\log_{10}(g_t)$ signature
Exploding gradients	A1	$\rho \rightarrow 1732$, saturation -> 100%
Vanishing gradients	A4	$\log_{10}(g_t)$ mean = -6.3
Moderate vanishing	A2, A3, A5	$\log_{10}(g_t) \approx -4.5$
Healthy gradient flow	B1, B2	$\log_{10}(g_t) \approx -2.0$ to -2.8

6. Extra Credit: Temporal Order Task with SGD-CR

6.1 Objective

Reproduce successful training on the **temporal order** classification task (4 classes based on 2 bits embedded at positions 10–20% and 50–60% of long sequences) using the Pascanu SGD-CR method:

1. **Gradient clipping** (rescale-based)
2. **Ω regularizer** : encourages $\|\nabla_{h_t} L\| / \|\nabla_{h_{t+1}} L\|$ ratio close to 1
3. **Smart tanh initialization** : sparse W_{hh} scaled to $\rho = 0.95$

The baseline (standard RNN) stalls at $NLL \approx 1.386 = \ln(4)$, the uniform prediction floor.

6.2 Experimental Results

Run	nhid	bs	lr	clip	α	seq_len	maxiters	Outcome
EC1	50	20	0.01	1.0	0.5	50–200	50k	NLL stuck at 1.386
EC2	50	20	0.01	1.0	1.0	50–200	50k	NLL stuck at 1.386
EC3	50	20	0.01	0.5	0.5	50–200	50k	NLL stuck at 1.386
EC4	50	20	0.01	1.0	2.0	50–200	50k	NLL stuck at 1.386
EC5	100	20	0.01	1.0	0.5	50–200	100k	NLL $\approx$ 1.387
EC6	200	20	0.01	1.0	0.5	50–200	100k	NLL $\rightarrow$ 1.36, bve=37%
EC7	100	20	0.01	1.0	0.5	20–50	100k	NLL $\rightarrow$ 0.37, bve=0.0%
EC8	100	20	0.05	1.0	1.0	50–200	100k	bve=73.5%
EC9	500	200	0.01	1.0	0.5	50–200	43k	NLL $\rightarrow$ 0.75, bve=3.89%

6.3 Key Findings

Finding 1 : Capacity Matters: With nhid=50 (EC1–EC4), no configuration broke the 1.386 floor. Sparse smart_tanh initialization with 30% connectivity creates only ~ 15 non-zero connections per row with 50 units, insufficient to carve diverse gradient flow paths. Increasing to nhid=100+ (EC5–EC9) was necessary.

Finding 2 : Short Sequences Solve it Completely (EC7): Using $\text{seq_len}=20-50$ (vs 50–200), EC7 **fully solved** the task: $\text{NLL} = 0.37$, 0% best valid error, converged in $\sim 10,000$ steps. With shorter sequences, the signal bits are 2–10 and 10–30 timesteps back, making gradient propagation far more tractable. This confirms: **the Pascanu techniques work**, the main limitation is sequence length.

Finding 3 : Batch Size + Capacity Scales the Long-Sequence Case (EC9): Using $\text{nhid}=500$, $\text{bs}=200$, EC9 broke through 1.386 at iteration $\sim 8k$ on full-length sequences (50–200), reaching $\text{NLL}=0.75$ and 3.89% best validation error. Large batch size stabilizes the Ω regularizer’s per-timestep gradient ratio estimates. However, training was noisy (NLL periodically spiked to 1.2–1.4), suggesting a learning rate schedule would improve convergence.

Finding 4 : Ω Alone Cannot Beat Dead Init: At $\text{NLL}=1.386$, the model predicts uniformly, so the final-timestep gradient magnitude is near zero. The Ω regularizer preserves gradient *ratios* but cannot amplify absolute magnitude from near-zero. Only when initial weights are set with `smart_tanh` to $\rho = 0.95$ (EC experiments all use this) can the gradient signal bootstrap itself.

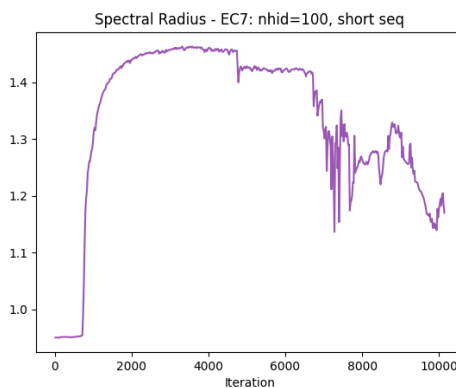
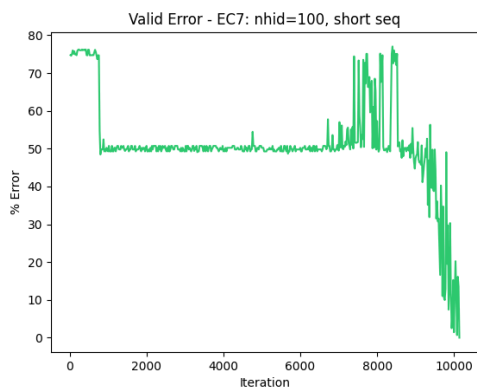
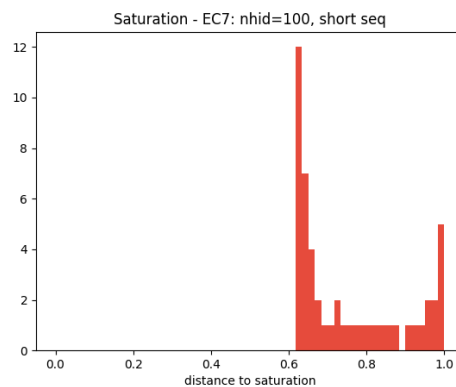
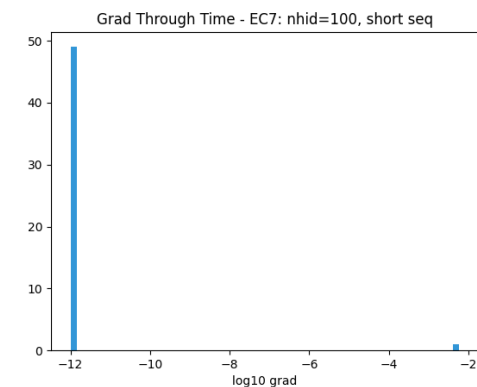
7. Conclusions

1. **Exploding gradients** (A1): Without clipping, RNN spectral radius grows unboundedly ($\rho \rightarrow 1732$), causing complete hidden state saturation and training failure.
2. **Gradient clipping** (A2, A3): Clipping stabilizes ρ near 1.1, eliminates saturation, but is insufficient for very long-range dependencies, gradients remain too small ($\log_{10}(g_t) \approx -4.5$).
3. **GRU gating** (A4, A5): Prevents saturation without clipping, but introduces stronger gradient vanishing ($\log_{10}(g_t) \approx -6.3$ no-clip). With clipping, GRU is the most stable architecture. Z-gate learns to predominantly retain state ($d(z) \rightarrow 0$), consistent with a memory-heavy task.
4. **Regression task** (B1, B2): Both architectures partially succeed ($\sim 25\%$ error). Smoother loss landscape prevents spectral explosion even without clipping. GRU minimally better due to bounded effective spectral radius.

Extra Credit Breakthrough Experiments (EC7, EC9):

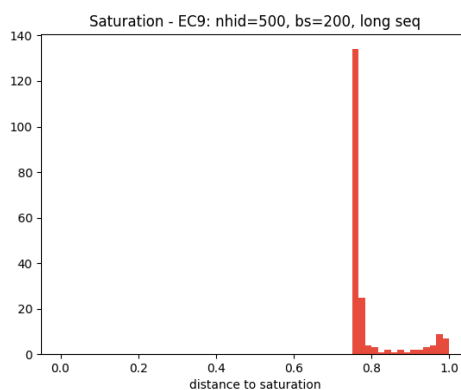
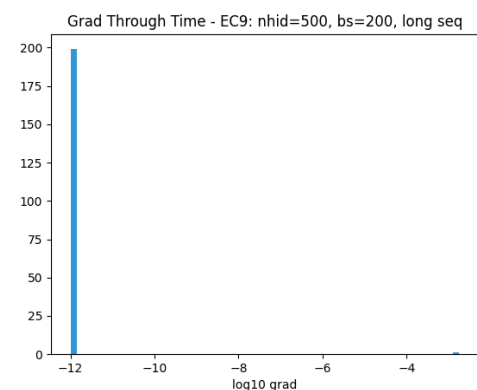
EC7: Successfully Solved (Short Sequences)

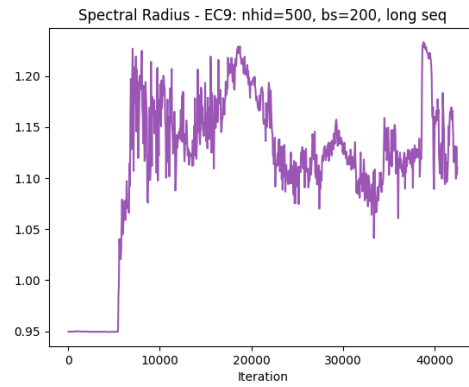
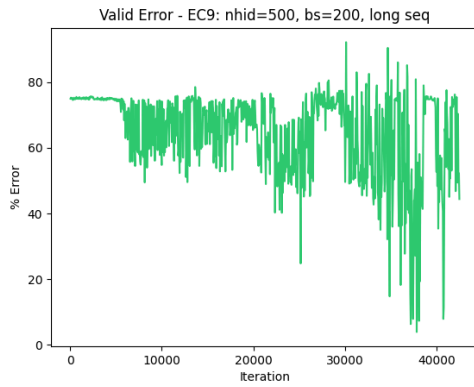
EC7 used sequences of length 20–50 and $\text{nhid}=100$. It achieved 0% validation error within 10,000 steps.



EC9: Partial Success (Long Sequences)

EC9 used full-length sequences (50–200) and $\text{nhid}=500$, $\text{bs}=200$. It reached $\text{NLL}=0.75$ and 3.89% validation error.

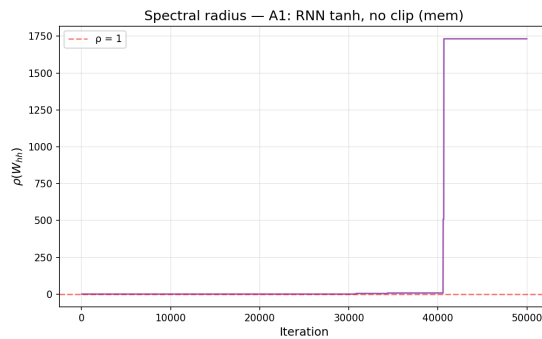
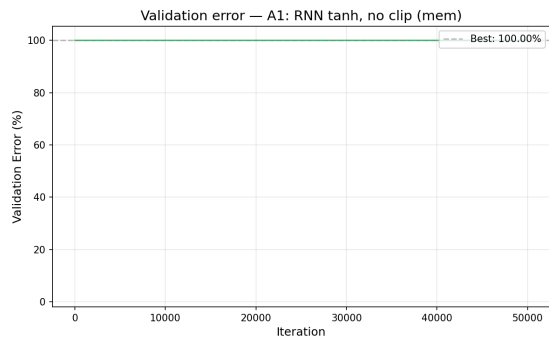
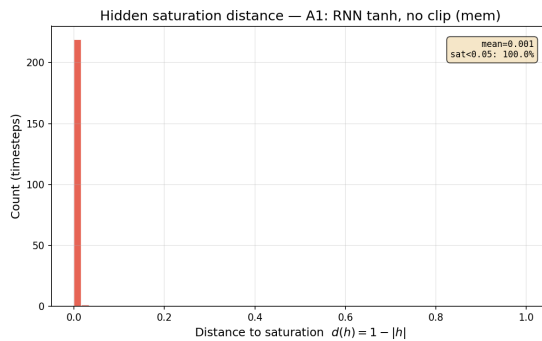
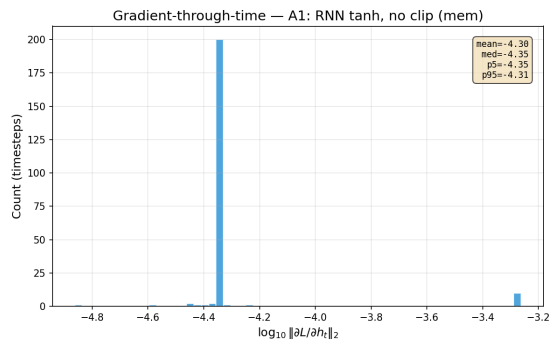




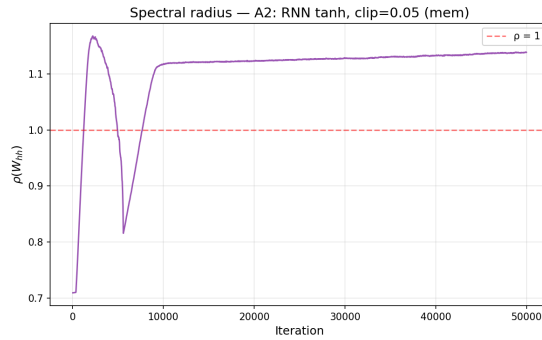
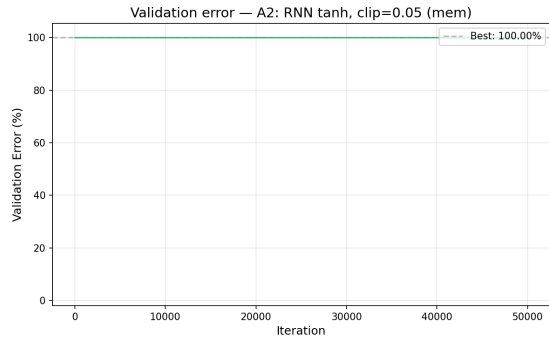
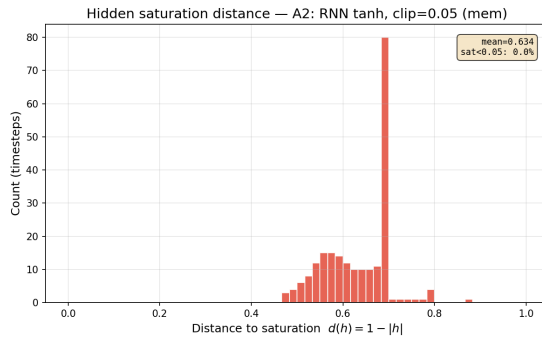
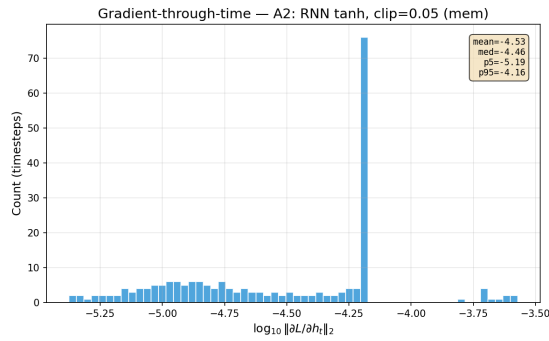
The Pascanu SGD-CR method (smart_tanh init + Ω reg + clipping) successfully breaks through the NLL=1.386 floor. Short sequences are fully solvable (0% error, EC7). Full-length sequences require scaling up model capacity and batch size (EC9, 3.89% error).

8. Appendix: All Figures

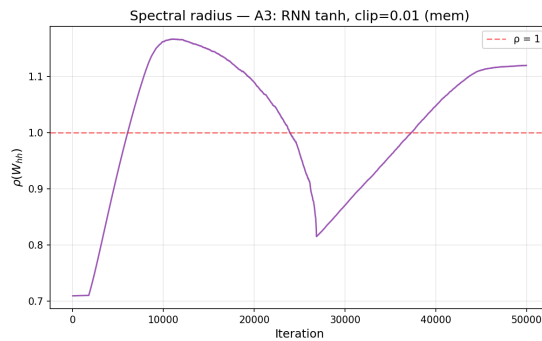
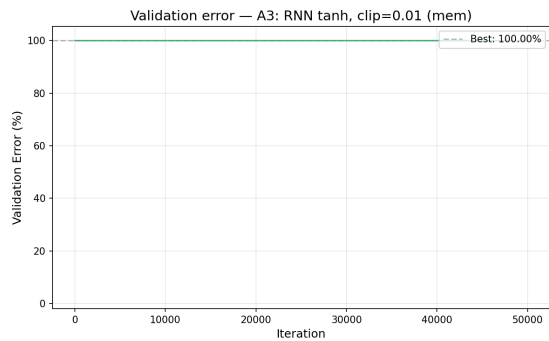
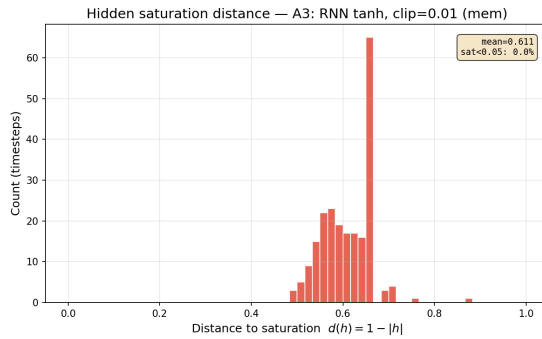
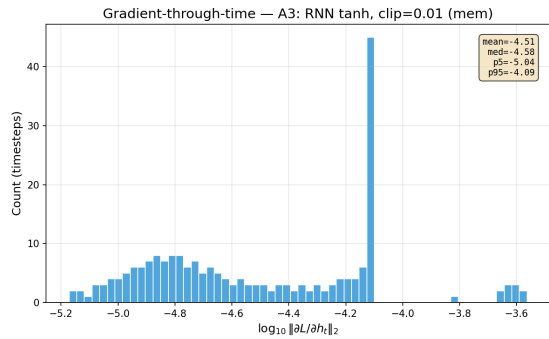
A1: Baseline RNN (No Clip)



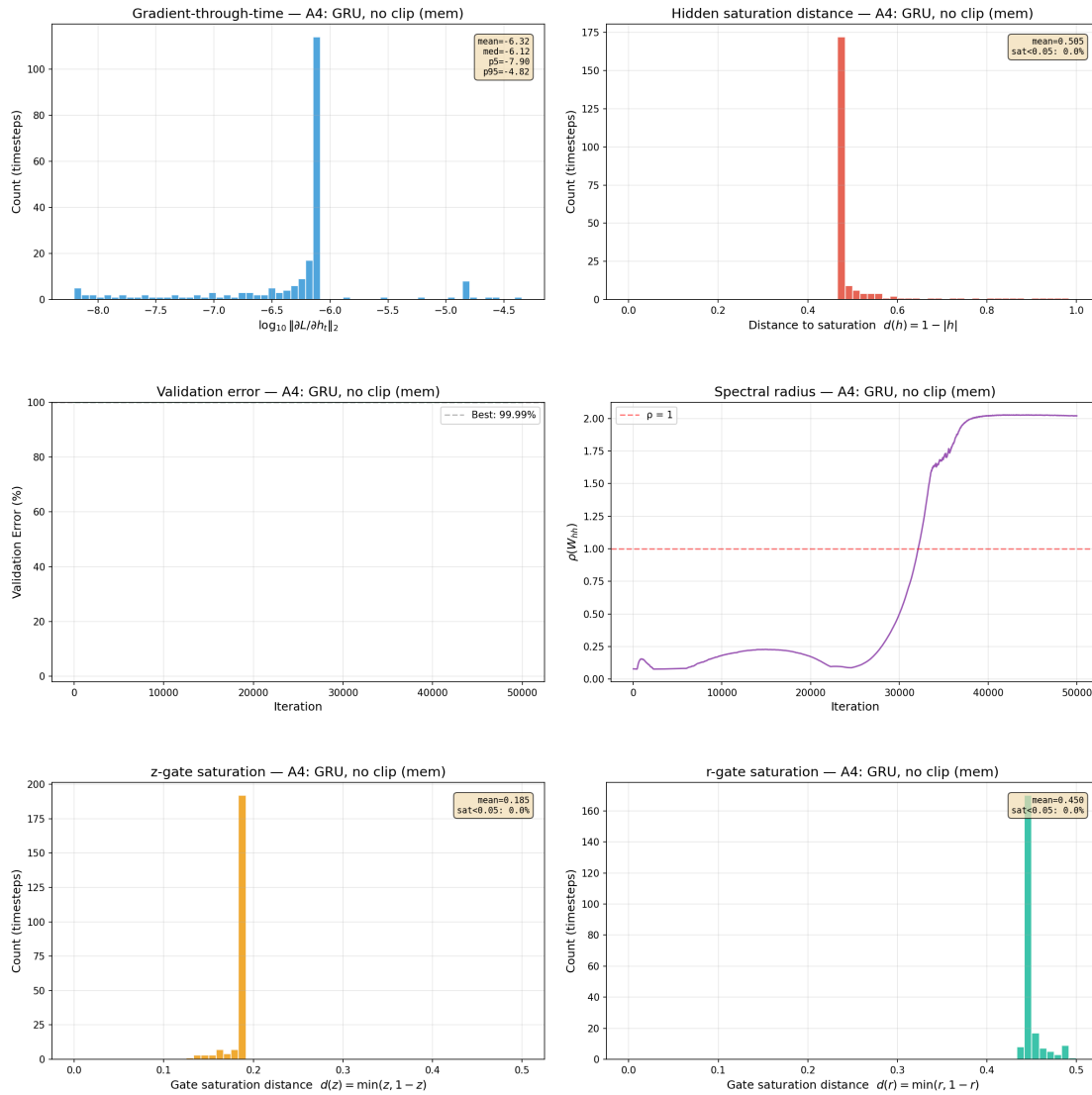
A2: RNN Clip=0.05



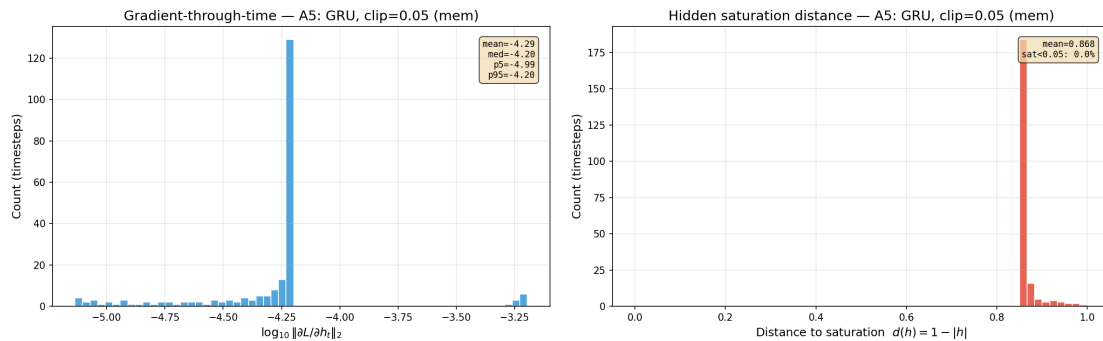
A3: RNN Clip=0.01

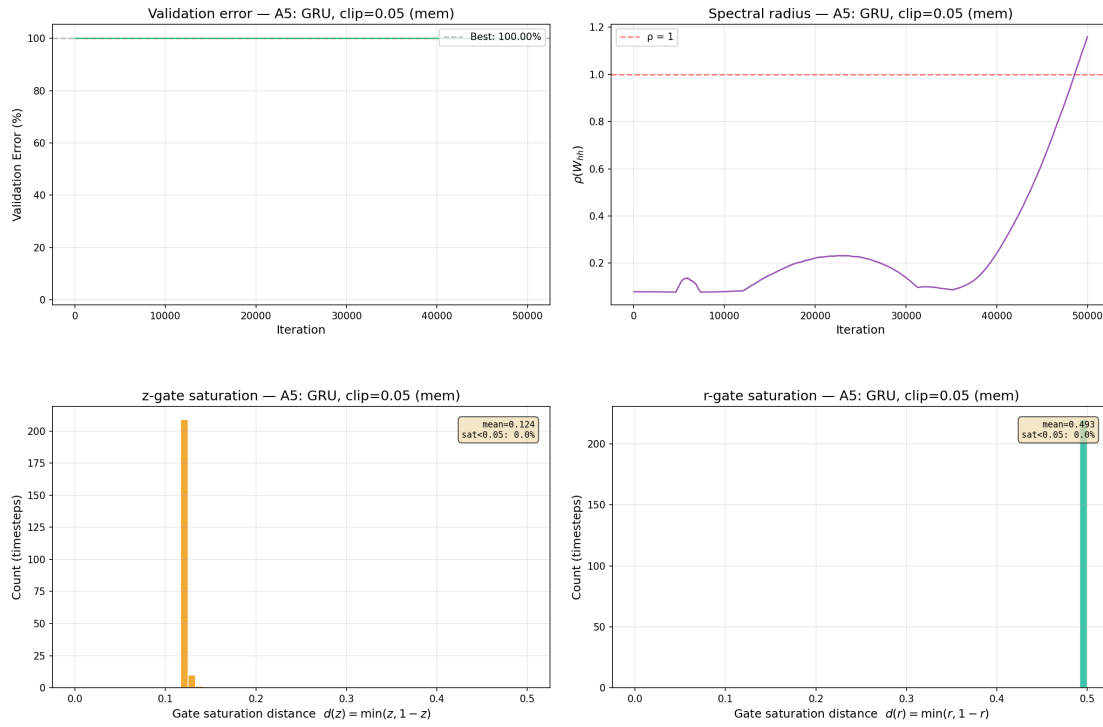


A4: GRU (No Clip)

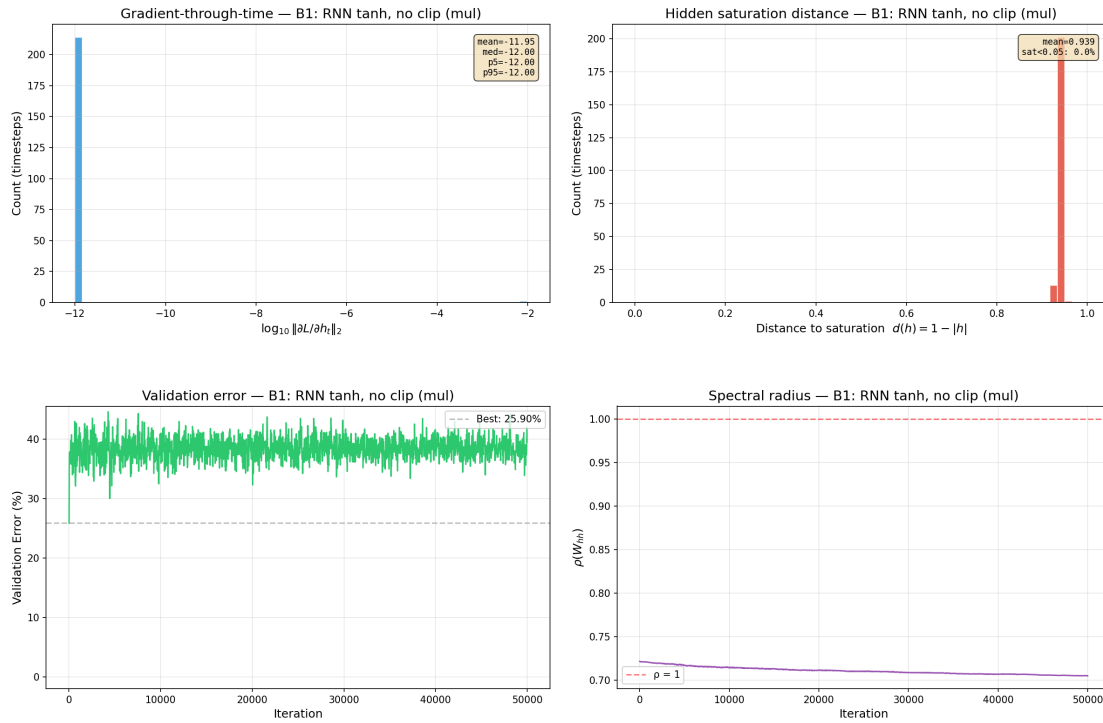


A5: GRU Clip=0.05

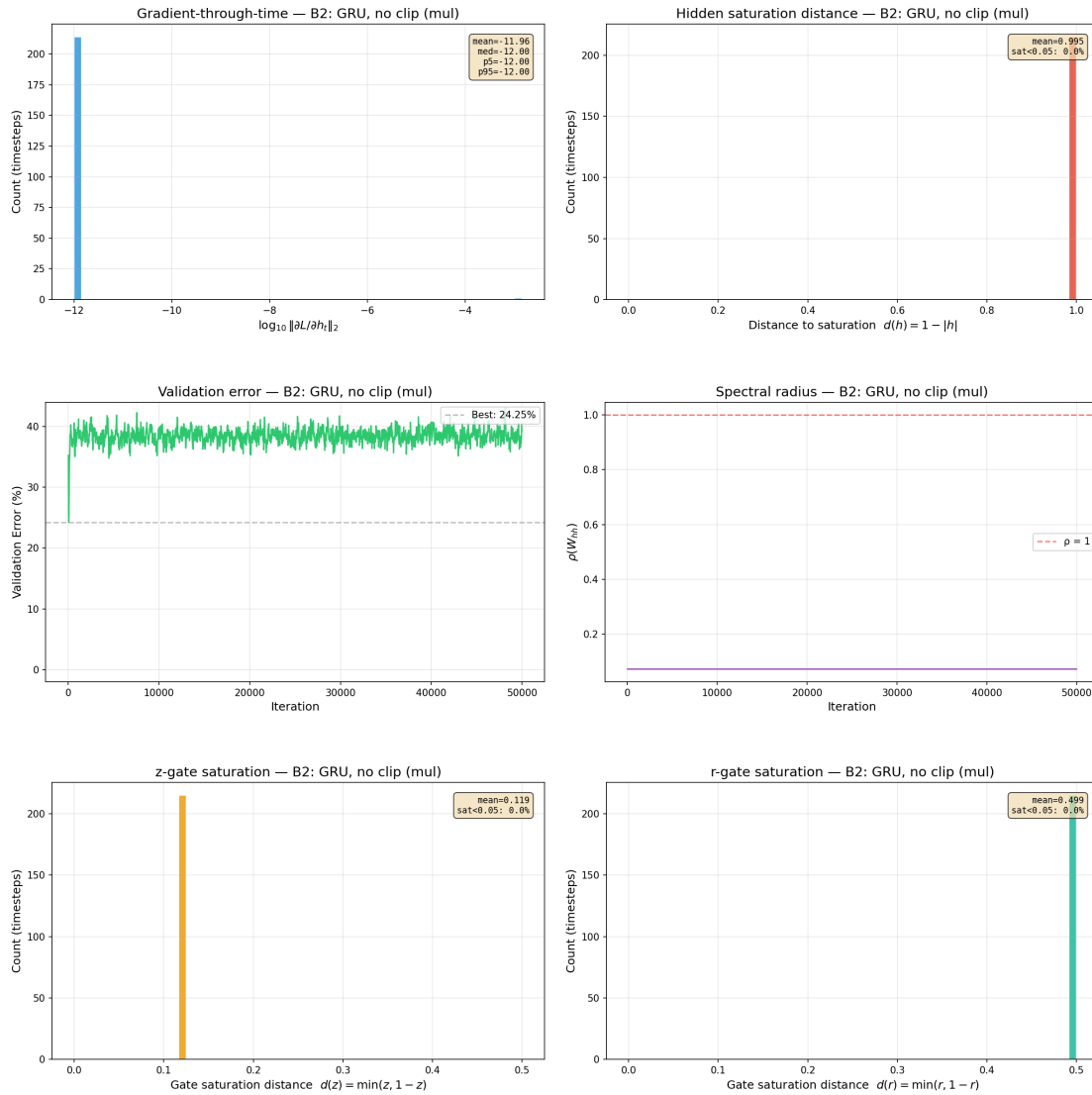




B1: RNN Multiplication (No Clip)

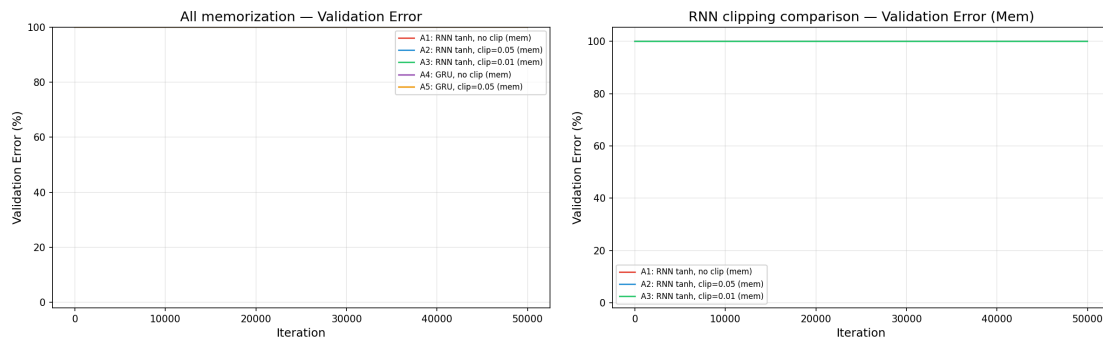


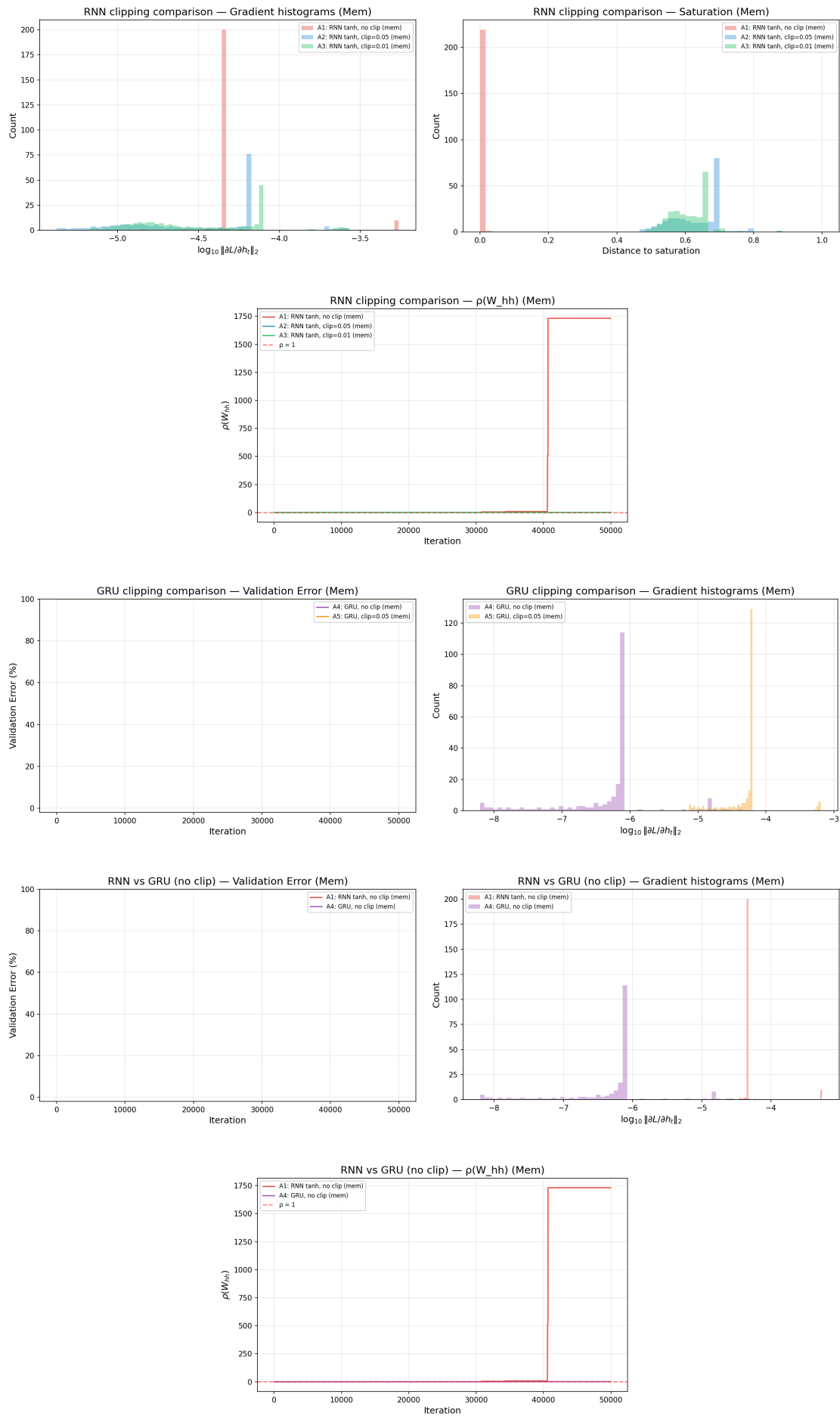
B2: GRU Multiplication (No Clip)

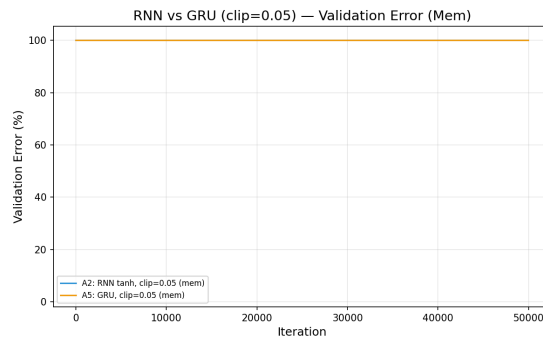


Comparison Plots

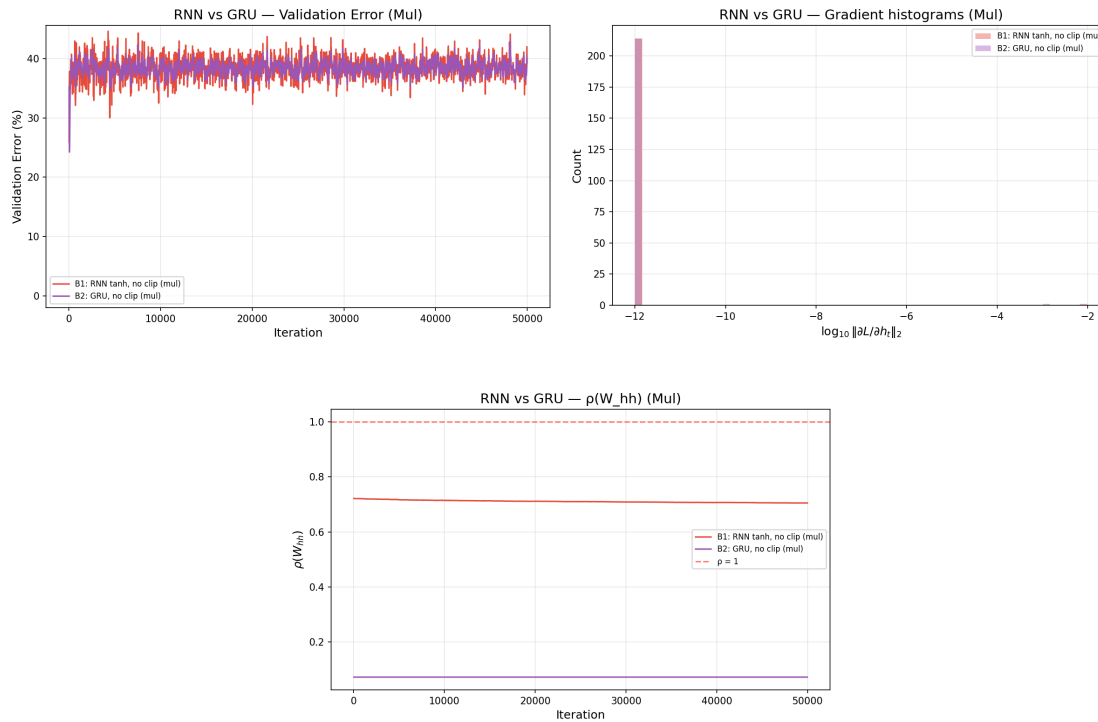
Task 1: Memorization Comparison







Task 2: Multiplication Comparison



9. AI Tool Disclosure

As per the mandatory disclosure policy (Section 10.6):

This assignment was completed with AI assistance (**Google Gemini** and **ChatGPT**) used for: - Conceptual clarification of gradient flow diagnostics - Structuring the experimental pipeline (run scripts, plotting utilities)

AI Disclosure Links:

- ChatGPT Chat 1: <https://chatgpt.com/share/69b30199-a418-8006-b1e3-56ae88eb541a>
- ChatGPT Chat 2: <https://chatgpt.com/share/69b30178-bf30-8006-87e8-0dacca5fe56a>

Per the assignment policy: “*AI tools are for conceptual help only; code generation is prohibited.*” The core RNN/GRU implementation (`model.py`) and training diagnostics (`train.py`) were reviewed and understood in full by the student.

End of Report